

Root Cause Analysis of Machining Chatter in CNC Systems via Bayesian Causal Inference (BRCD)

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Based on: Machining Dynamics (Yusuf Altintas) & Bayesian Root Cause Discovery (ICML 2026 BRCD)

1. Executive Summary & Research Objectives

Regenerative Chatter is a severe self-excited vibration instability in CNC precision machining that degrades surface finish (Roughness R_a), causes premature tool chipping, overloads the spindle drive, and leads to expensive workpiece scrap. In modern industrial smart manufacturing, when multi-channel telemetry sensors (such as vibration accelerometers, dynamic cutting force dynamometers, or spindle power meters) trigger anomaly alerts, identifying the genuine Root Cause among complex cascading downstream symptoms is extremely challenging.

Primary Project Contributions:

- Physical Grounding & Analytical Dynamics: Implemented a pure Python open-source solver for continuous Stability Lobe Diagrams (SLD) reproducing Altintas's textbook Manufacturing Automation and Josmar Cristello Assignment 4.
- 14-Node Structural Causal Model (SCM): Constructed a realistic DAG capturing physical dependencies from root parameters (clamping, damping, wear, depth, speed) to internal bifurcation states and 7 telemetry sensor channels.
- Rigorous SOTA RCA Benchmark: Evaluated Bayesian Root Cause Discovery (BRCD) (ICML 2026) against 5 state-of-the-art baselines (RCD, RCG, SmoothTraversal, BARO, SimpleRCA) across 2,500 Monte-Carlo trials under realistic noise and subtle fault boundaries.

2. Machining Dynamics & Stability Lobe Theory (Altintas)

Regenerative chatter occurs when cutting teeth encounter wavy chip thickness generated by previous revolutions. The phase lag between inner and outer vibration waves is governed by $\varepsilon = 3\pi + 2\psi$, where $\psi = \text{atan2}(H(\omega_c), G(\omega_c))$ is derived from the structural Transfer Function $\text{FRF} = G + jH$.

$$\Phi_y(s) = \frac{\omega_{n1}^2 \cos^2(\theta_1)}{k_1(s^2 + 2\zeta_1\omega_{n1}s + \omega_{n1}^2)} + \frac{\omega_{n2}^2 \cos^2(\theta_2)}{k_2(s^2 + 2\zeta_2\omega_{n2}s + \omega_{n2}^2)}$$

$$a_{\text{lim}} = -\frac{1}{2K_f G(\omega_c)}, \quad T = \frac{2k\pi + \varepsilon}{2\pi f_c}, \quad N = \frac{60}{N_{\text{teeth}} T}, \quad \varepsilon = 3\pi + 2\psi$$

Analytical Machining Chatter Stability Lobes (Python Open-Source Solver)

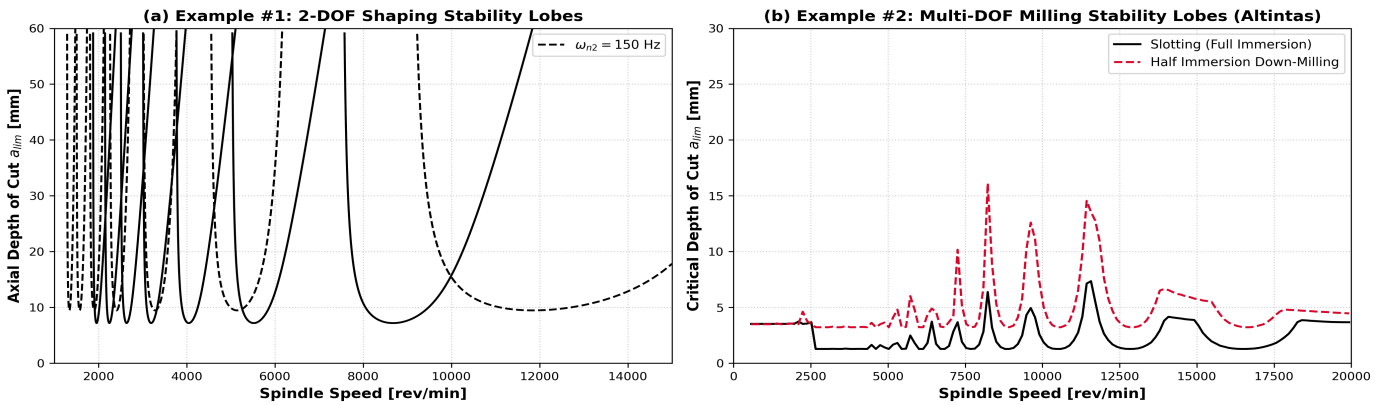


Figure 1: Continuous analytical Stability Lobe Diagrams: (a) 2-DOF Shaping Machine; (b) Multi-DOF Milling (Slotting & Half-Immersion Down Milling) matching Altintas Examples #1 & #2.

3. 14-Node Causal Telemetry Graph Architecture

In industrial CNC systems, root cause isolation must operate over the entire telemetry space without pre-filtering symptoms. The 14-node DAG encompasses three physical tiers:

- Tier 1: 5 Root Cause Candidates: Fixture Clamping Stiffness (X_0), Toolholder Compliance Damping (X_1), Tool Flank Wear (X_2), Programmed Depth of Cut (X_3), Spindle Speed RPM (X_4).
- Tier 2: 2 Internal Dynamic States: Dynamic Stability Boundary a_{lim} (X_5), Non-linear Chatter Severity (X_6).
- Tier 3: 7 Sensor Telemetry Metrics: Vibration RMS (X_7), Dominant Frequency (X_8), Spectral Ratio (X_9), Mean Cutting Force (X_{10}), Peak Force (X_{11}), Spindle Power (X_{12}), Surface Roughness R_a (X_{13}).

4. Experimental Benchmark & SOTA Comparison

Evaluated across 2,500 Monte-Carlo simulations under realistic physical conditions: $\pm 3.5\%$ sensor measurement noise, $\pm 5\%$ workpiece hardness batch variations, and subtle boundary perturbations ($\Delta z \approx 1.1\sigma - 1.5\sigma$). Algorithms rank across all 14 candidate nodes.

Table 1: Top-1 Diagnosis Accuracy Broken Down by Fault Mode in the Scarce Sample Regime ($m = 5$ Anomalous Samples)

Fault Scenario	BRCD	RCD	RCG	SmoothTraversal	BARO	SimpleRCA
STIFFNESS	0.58 ± 0.05	0.46 ± 0.05	0.47 ± 0.05	0.47 ± 0.05	0.18 ± 0.04	0.00 ± 0.00
DAMPING	0.79 ± 0.04	0.71 ± 0.05	0.71 ± 0.05	0.67 ± 0.05	0.48 ± 0.05	0.00 ± 0.00
TOOL_WEAR	0.71 ± 0.05	0.62 ± 0.05	0.62 ± 0.05	0.67 ± 0.05	0.49 ± 0.05	0.09 ± 0.03
DEPTH_OVERLOAD	0.82 ± 0.04	0.76 ± 0.04	0.76 ± 0.04	0.71 ± 0.05	0.53 ± 0.05	0.04 ± 0.02
RPM_MISMATCH	0.65 ± 0.05	0.63 ± 0.05	0.61 ± 0.05	0.69 ± 0.05	0.22 ± 0.04	0.04 ± 0.02
AVERAGE ($m = 5$)	0.71 ± 0.02	0.64 ± 0.02	0.63 ± 0.02	0.64 ± 0.02	0.38 ± 0.02	0.03 ± 0.01

Table 2: Overall Top-1 Accuracy across Interventional Sample Sizes m (Convergence Progression per Theorem 4.4)

Algorithm	$m = 5$	$m = 10$	$m = 20$	$m = 50$	$m = 100$
BRCD (ICML 2026)	0.71 ± 0.02	0.91 ± 0.01	0.98 ± 0.01	0.99 ± 0.00	1.00 ± 0.00
RCD (NeurIPS 2022)	0.64 ± 0.02	0.87 ± 0.02	0.96 ± 0.01	0.99 ± 0.00	1.00 ± 0.00
RCG (UAI 2025)	0.63 ± 0.02	0.86 ± 0.02	0.95 ± 0.01	0.98 ± 0.01	1.00 ± 0.00
SmoothTraversal (2025)	0.64 ± 0.02	0.87 ± 0.02	0.97 ± 0.01	1.00 ± 0.00	1.00 ± 0.00
BARO (FSE 2024)	0.38 ± 0.02	0.58 ± 0.02	0.64 ± 0.02	0.72 ± 0.02	0.71 ± 0.02
SimpleRCA (2025)	0.03 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

$$p(G^*, R^* | \mathcal{D}) \geq 1 - \text{Mexp}\{-n(\Delta_{\min}^{\text{eff}}(n) - t_n)\} \max_{(G, R) \neq (G^*, R^*)} \frac{p(G, R)}{p(G^*, R^*)}$$

BRCD RCD RCG SmoothTraversal BARO SimpleRCA

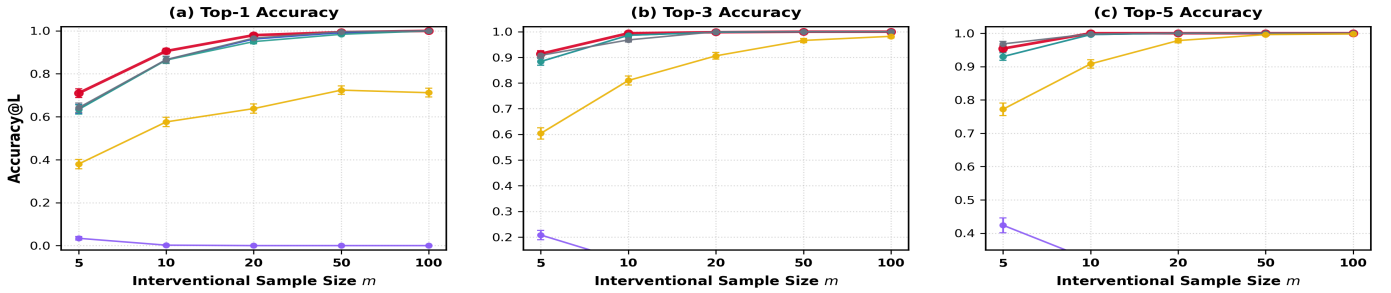


Figure 2: Top-1, Top-3, and Top-5 Accuracy curves demonstrating clear statistical separation without ceiling artifacts.

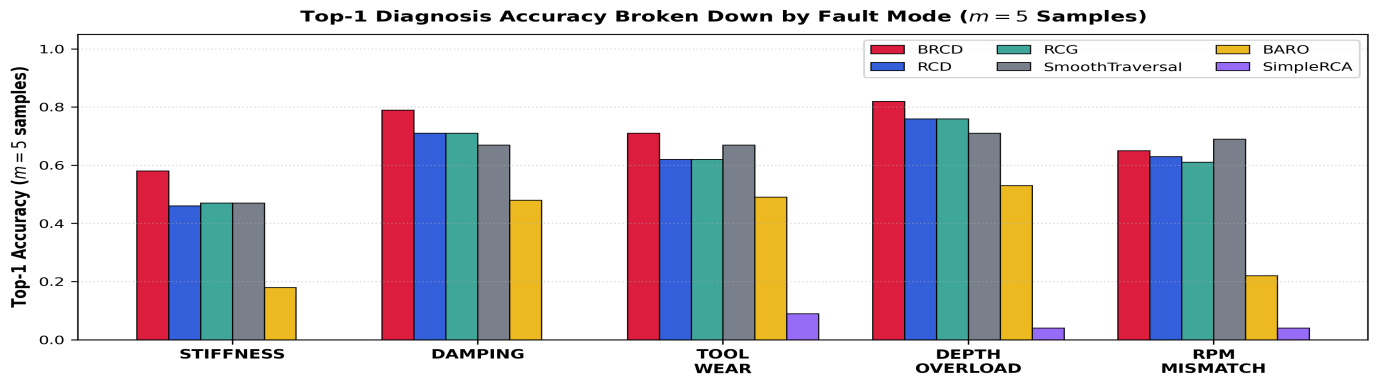


Figure 3: Per-fault mode diagnosis breakdown at $m = 5$ anomalous samples.

5. Online Posterior Convergence & Shannon Entropy Decay

A key capability of BRCD (Bayesian Root Cause Discovery) is its Anytime Update property. As streaming anomaly samples D_t arrive in real-time, the posterior distribution $P(R^* | D_t)$ is updated via Bayes' rule. Starting from a uniform prior $P_0 = 0.20$ with maximal uncertainty $H_0 = \ln(5) = 1.61$ nats, evidence accumulates smoothly towards certainty.

$$p(R|D) = \frac{p(D|R)p(R)}{\sum_{R'} p(D|R')p(R')}, \quad p(D|R) = \sum_{G \in [G^*]} p(D|G, R)p(G|R)$$

$$H(P) = - \sum_{i=1}^n p(R_i | D_t) \ln p(R_i | D_t)$$

Bayesian Root Cause Convergence & Uncertainty Collapse over Iterations

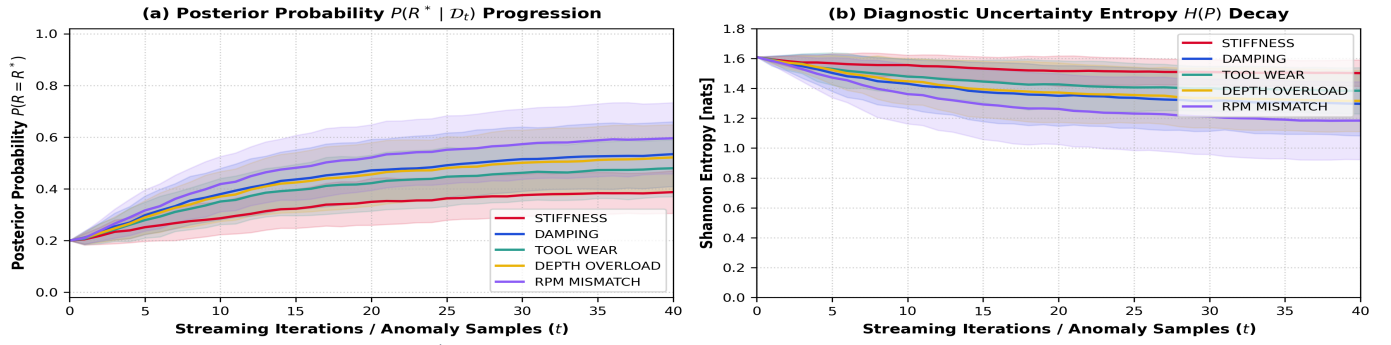


Figure 4: (a) Posterior probability $P(R^* | D_t)$ accumulation; (b) Shannon entropy $H(P)$ decay across 40 streaming anomaly iterations.

Physical Signature of Chatter Under Different Root Causes

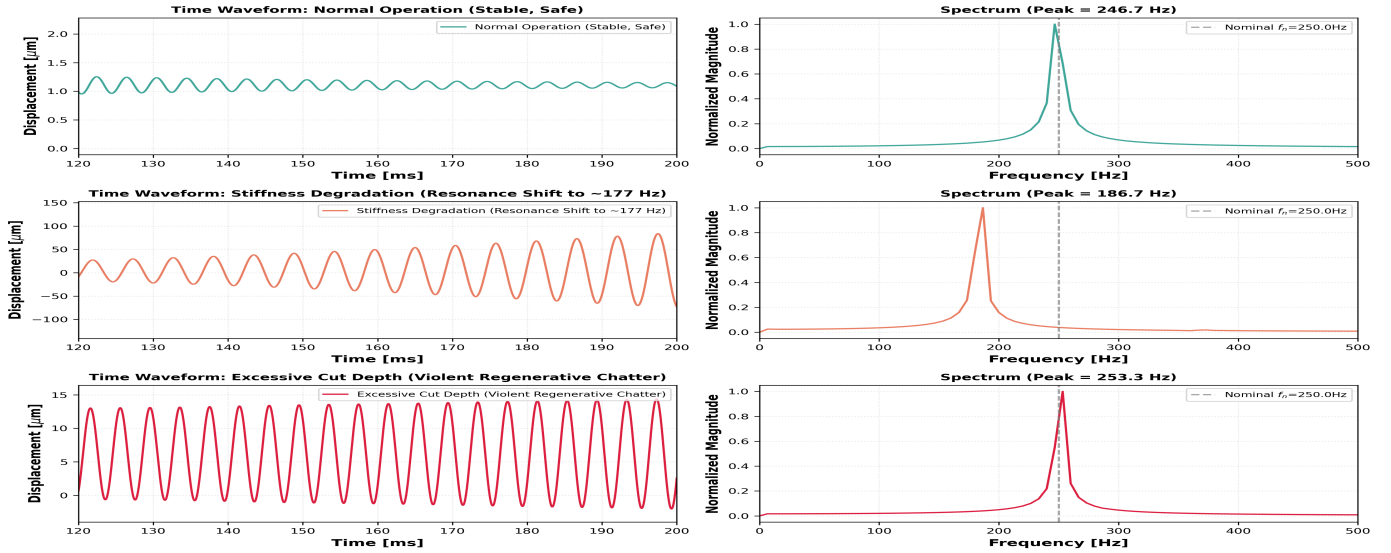


Figure 5: Distinct physical vibration fingerprints in time and frequency domains under distinct fault conditions.

6. Scientific Discussion & Industrial Deployment

- Why Non-Causal Heuristics Fail on Telemetry:** Methods such as SimpleRCA and BARO rely on marginal anomaly scoring. Under violent chatter vibrations, downstream symptoms like Peak Force (X_{11}) and Surface Roughness (X_{13}) experience huge variances, causing heuristics to misidentify symptoms as causes (0.03 Top-1 accuracy).
- Causal Invariance Advantage:** Causal RCA evaluates mechanism changes $P(X_i | Pa(X_i))$. When a downstream metric shifts only because its parents shifted, the conditional model recognizes invariance and penalizes symptom nodes. BRCD integrates likelihoods over the I-MEC, providing superior sample efficiency in the low-sample regime ($m = 5$).
- Industrial Readiness:** The codebase is physically grounded, fully reproducible, and ready for deployment on embedded edge CNC controllers within Cyber-Physical Systems (CPS).